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Cascading StyleSheets

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What are Cascading StyleSheets?

- CSS is an extension to basic HTML that allows you to style elements within your web pages.
- CSS can be embedded into your webpage or store in an external text file that you can link to your webpage.
- Control design aspects of an HTML document.
- Contain rules that define appearance of elements.

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Ways of Using CSS

- Inline styles.
- Embedded/Internal style sheets.
- External style sheets.

Syntax

- CSS rules are composed of three parts:
  - Selector.
  - Property.
  - Value.

- Template:

```
selector { property:value; property:value;}
```

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Syntax

- Example – External/Embedded CSS:

```
h1{
        font-style: italic;
        color: blue;
    }
```

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Syntax

- Example – Inline:

```
<h1 style =“font-style: italic;
color: blue;”></h1>
```

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CSS Rule Examples

```
h1{ font-style: italic; color: blue; }

h2{ font-style: italic; color: blue; }

h3{ font-style: italic; color: blue; }

h1,h2,h3{
      font-style: italic;
      color: blue;
}
```

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CSS Selectors

ID

- You can customise and create styles for a unique element within your webpage through the use of ID’s.

- You specify an ID for an element within the HTML:

test.html test.css

```html
<!DOCTYPE html>
<html>
<head>
    <link rel="stylesheet" type="text/css" href="test.css">
    <title>Test Page</title>
</head>
<body>
    <p id="testID">Test Paragraph</p>
</body>
</html>
```

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CSS Selectors

ID

- You can then use the hash (#) notation to apply style rules to that element within your CSS:

```
test.html    test.css
#testID {
    font-size: 16px;
}
```

- This should only be used when you have a single element that you want to apply styling rules upon.

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CSS Selectors

Class

- If you wish to create common styles that can be applied to multiple elements within your HTML, then you can use the class feature.

- You can specify a class within your HTML:

test.html × test.css ×

```html
<!DOCTYPE html>
<html>
<head>
    <link rel="stylesheet" type="text/css" href="test.css">
    <title>Test Page</title>
</head>
<body>
    <p class="paragraphStyle">Test Paragraph</p>
</body>
</html>
```

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CSS Selectors

Class

- You can then use the dot (.) notation to apply style rules to that element within your CSS:

- This should only be used when you have common styling that you want to applied to multiple elements within your website.

CSS Selectors

Element

- You can also apply styles to all items of a certain element (or tag) type:
  - E.g. All paragraphs .
- This is done by using the tag type within the CSS:

test.html test.css

```html
<!DOCTYPE html>
<html>
<head>
    <link rel="stylesheet" type="text/css" href="test.css">
    <title>Test Page</title>
</head>
<body>
    <p>Test Paragraph</p>
</body>
</html>
```

test.html test.css

```css
p {
    font-size: 16px;
}
```

CSS Selectors

Element Subtypes

- You can also customise subtypes of elements, defining different styles for the same element or group.
- Defined within an embedded or external style sheets.
- Syntax:

```
selector.classname {
        property: value;
        property: value;
}
<selector class="classname">
```

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CSS Selectors

Element Subtypes

- You can also customise subtypes of elements, a common use of this is for styling hyperlinks:

```
a:link {text-decoration: underline
                    overline; color: red;}

a:active {text-decoration: none; color:
                    yellow;}

a:visited  {text-decoration: overline;}

a:hover {text-decoration: none;
    background: green; color: red;}
```

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CSS Selectors

Element Subtypes

```
h1.go {color: green;}
h1.stop { color: red;}
h1.caution{ color: yellow; }
p{ font-family: sans-serif; color: blue; }
```

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CSS Selectors

Element Subtypes

```
<body>
   <h1 class="go">Paragraph One</h1>
   <p>This is a basic HTML document that has three
   paragraphs. They may be simple, but I think they will
   get the point across. The font for each paragraph is
   scans-serif and the colour is blue. This is the first
   paragraph, which has a level one heading that is blue
   and with the border of thick groove. </p>
   <h1 class="caution">Paragraph Two</h1>
   <p>This is the second paragraph in our embedded style
   adventure. It has a level two heading that is green and
   italic.</p>
   <h1 class="stop"> Paragraph Three</h1>
   <p>This is the third paragraph in our embedded style
   adventure. It has a level six heading that is aqua.</p>
</body>
</html>
```

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CSS Selectors

Element Subtypes

Paragraph One

This is a basic HTML document that has three paragraphs. They may be simple, but I think they will get the point across. The font for each paragrah is scans-serif and the colour is blue. This is the first paragraph, which has a level one heading that is blue and with the border of thick groove.

Paragraph Two

This is the second paragraph in our embeded style adventure. It has a level two heading that is green and italic.

Paragraph Three

This is the third paragraph in our embeded style adventure. It has a level six heading that is aqua.

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CSS Methods

Inline CSS

- Allows for brief deviation of other defined styles.
- Adding the style attribute to an HTML element.

<element style="declaration">

- Does **not** let us achieve the separation of style from the document.

test.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Test Page</title>
</head>
<body>
    <p style="font-size: 16px;">Test Paragraph</p>
</body>
</html>
```

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CSS Methods

Inline

- The element can be used to apply very localised styling within an element.

```
<p>This is a basic HTML document that uses the
span element to change the font size of the
second sentence.
<span style="font-size:200%;">It may be simple,
but I think you will get the idea.</span>
The font size is now back to normal for the
remainder of the paragraph.
</p>
```

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CSS Methods

Inline

This is a basic HTML document that uses the span element to change the font size of the second sentence. It may be simple, but I think you will get the idea. The font size is now back to normal for the remainder of the paragraph.

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CSS Methods

Embedded/Internal CSS

- Placed within the <head> section of document.

- Embedded Style Syntax:

```
<style type="text/css">
<!--
      The style rules are placed here
-->
</style>
```

test.html

```
<!DOCTYPE html>
<html>
<head>
    <style type="text/css">
        #testID {
            font-size: 16px;
        }
    </style>
    <title>Test Page</title>
</head>
<body>
    <p id="testID">Test Paragraph</p>
</body>
</html>
```

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CSS Methods

Embedded/Internal CSS

```
<style type="text/css">
<!--
  h1{ color: red; border: thick groove;}
  h2{ font-style: italic; color: green; }
  h6{ color: aqua; }
   p{ font-family: sans-serif; color: blue; }
 -->
</style>
```

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CSS Methods

Embedded/Internal CSS

Paragraph One

This is a basic HTML document that has three paragraphs. They may be simple, but I think they will get the point across. The font for each paragrah is scans-serif and the colour is blue. This is the first paragraph, which has a level one heading that is blue and with the border of thick groove.

Paragraph Two

This is the second paragraph in our embeded style adventure. It has a level two heading that is green and italic.

Paragraph Three

This is the third paragraph in our embeded style adventure. It has a level six heading that is aqua.

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CSS Methods

External CSS

- Also known as Linked Style Sheets.
- Stored separately from the HTML document.
- Contains the same rules.
- Can be used to apply the same style to every page in a Web site

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CSS Methods

External Style Sheets

- <link> element is used to apply an external style sheet to an HTML document.
  - href attribute is used to provide the name and location of the style sheet.
  - rel attribute is used to specify the relationship of the page containing the link to the other document.
  - type attribute is used to tell the browser what type of file it is reading.

- Basic syntax for using the <link> element is:

```
<link rel="Stylesheet"  href=exstyle.css type="text/css" />
 Or
 <link rel="Stylesheet"  href="url"  type="text/css" />
```

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CSS Methods

External CSS

test.html test.css

```html
<!DOCTYPE html>
<html>
<head>
    <link rel="stylesheet" type="text/css" href="test.css">
    <title>Test Page</title>
</head>
<body>
    <p id="testID">Test Paragraph</p>
</body>
</html>
```

test.html test.css

```css
#testID {
    font-size: 16px;
}
```

Note: Do not add a space between the property value and the unit (such as `margin-left: 20 px;`). The correct way is: `margin-left: 20px;`

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The Cascade

Specificity

CSS Order

- It is possible to have an external CSS, and embedded CSS, and inline CSS all within the one HTML page.

- In this case there is a strict priority order:
  1. Inline
  2. Embedded
  3. External

- The most specific rules take precedence.

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Specificity

Selectors

- There is further ordering within individual style sheets:
  1. Type selectors (e.g. p)
  2. Class selectors (e.g. .genericParagraph)
  3. ID selectors (e.g. #paragraphOne)

- Universal selector (*), combinators (+, >, ||, etc.) have no effect on specificity.

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Specificity

Equal Specificity

- If the same rule is written twice within a CSS page but with different styling then lower rule will be selected – known as Latest Rule Counts.

```
h1 {background-color: yellow;}
h1 {background-color: red;}
```

- Red will be applied to h1 tags in this example.

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!important

- When the **!important** rule is used the declaration overrides any other declarations.
- However, using !important is considered **bad practice** and should be avoided:
  - It makes debugging difficult as it breaks the natural cascade.
  - When two conflicting declarations with the !important rule are applied to the same element, the declaration with a greater specificity will apply.

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!important

Rules of Thumb

- If you feel you need to use it go through the following points before considering using !important:
  1. **Always** look for a way to use specificity before even considering !important.
  2. **Only** use !important on page specific CSS that overrides foreign CSS.
  3. **Never** use !important when you're writing a plugin/mashup.
  4. **Never** use !important on site-wide CSS.

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Properties

Text Colour

- Syntax options:
  - color: #000000
  - color: black
  - color: rgb(0, 0, 0)
- Colour names:
  - 147 predefined names.
- RGB Values:
  - Values for red, green, and blue – rgb(R, G, B).
- Hexadecimal:
  - Six or three digit alpha-numerical - #FFFFFF, #000

```
body {
  color: blue;
}

h1 {
  color: green;
}
```

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Font-Family

- Fall Back – a comma separated list of fonts to use if the system doesn't have the font installed.

```
font-family: "Times New Roman", Times, serif;
```

- Quotes – use quotes when a font is multiple words, like "Times New Roman".

- End With Generic – End with a generic font type, like serif or san-serif. This way the browser will pick for you.

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Font Size

- Pixels – pixels are considered a fixed font size. Fonts will appear at this size relative to the screen.
- Percentages – percentages are relative from the default size of the text. 100% means the default text size.
- EM – EM is another way of doing relative measurement. Varies only slightly from percentage.
  - EM is equal to the current font size. So if the font size of the document is 12pt then 1em = 12pt, 2em = 24pt, etc.

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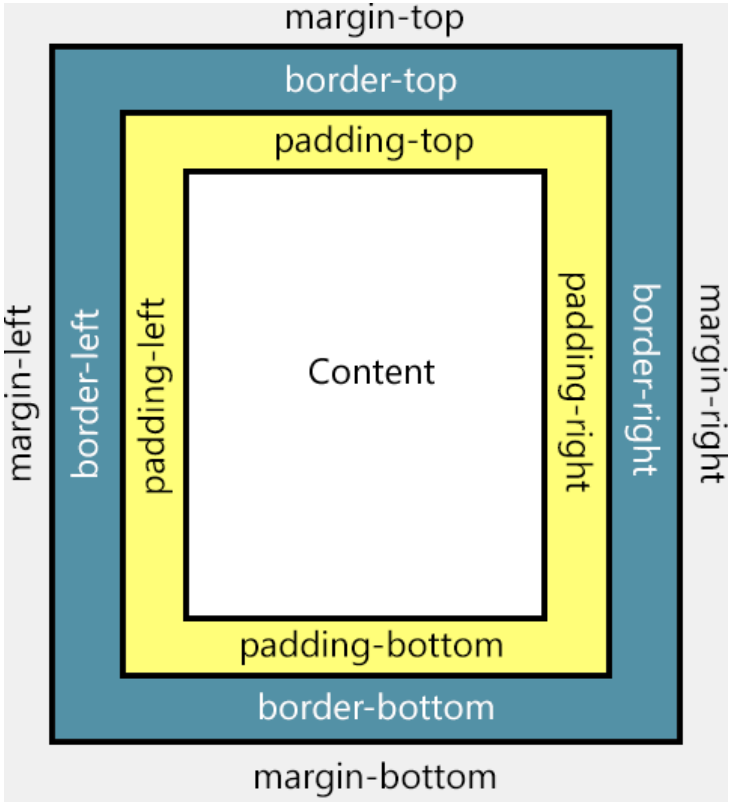
Border, Margin, and Padding

- Syntax:
  - margin-top: 10px;
  - margin-right: 10px;
  - margin-bottom: 10px;
  - margin-left: 10px;
- Short version:
  - Margin: 5px 10px 5px 10px;



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Borders

- Border-color, either specific sides or all:
  - border-color: #FFFFFF
  - border-top-color: #FFFFFF
- Border-width allows you to specify the thickness, either on a single side or for all sides:
  - border-width: 10px;
  - border-top-width: 10px;
- Border-style allows you to pick from a range of pre-defined styles:
  - solid, dotted, dashed, groove….

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Layout

Inline vs. Block vs. Inline-Block

- Every HTML element has a default display value depending on what type of element it is – the default display value for most elements is block or inline.

- Block – A block element always starts on a new line and takes up the full width available.

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Layout

Inline vs. Block vs. Inline-Block

- Inline – An inline element does not start on a new line and only takes up as much width as necessary.

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Layout

Inline vs. Block vs. Inline-Block

- Inline-block:
  - Major difference compared to inline is that it allows a width and height to be set.
  - Major difference compared to block is that it does not add a line-break after the element.

Layout

Inline vs. Block vs. Inline-Block

```
<style>
span.a {
  display: inline; /* the default for span */
  width: 100px;
  height: 100px;
  padding: 5px;
  border: 1px solid blue;
  background-color: yellow;
}

span.b {
  display: inline-block;
  width: 100px;
  height: 100px;
  padding: 5px;
  border: 1px solid blue;
  background-color: yellow;
}

span.c {
  display: block;
  width: 100px;
  height: 100px;
  padding: 5px;
  border: 1px solid blue;
  background-color: yellow;
}
</style>
```

display: inline

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vestibulum consequat scelerisque elit sit amet consequat. Aliquam erat volutpat. Aliquam venenatis gravida nisl sit amet facilisis. Nullam cursus fermentum velit sed laoreet.

display: inline-block

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vestibulum consequat scelerisque elit sit amet consequat. Aliquam erat volutpat. Aliquam venenatis gravida nisl sit amet facilisis. Nullam cursus fermentum velit sed laoreet.

display: block

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vestibulum consequat scelerisque elit sit amet consequat. Aliquam erat volutpat.

Aliquam

venenatis

gravida nisl sit amet facilisis. Nullam cursus fermentum velit sed laoreet.

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Visibility

- The visibility property defines whether or not a property is visible or not.
  - NB: hidden elements still take up space on the page. (use display:none if you want the space removed).

```
h2.a {
  visibility: visible;
}

h2.b {
  visibility: hidden;
}
```

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Questions?